Project: Worked examples
Title: Plate bending
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1. Scope and method

Calculates the deflection, von Mises stress and fundamental frequency of a thin, flat, isotropic plate under a lateral load, by Kirchhoff small-deflection theory. It covers circular and rectangular plates, simply supported or clamped, under uniform, patch or ring loads.

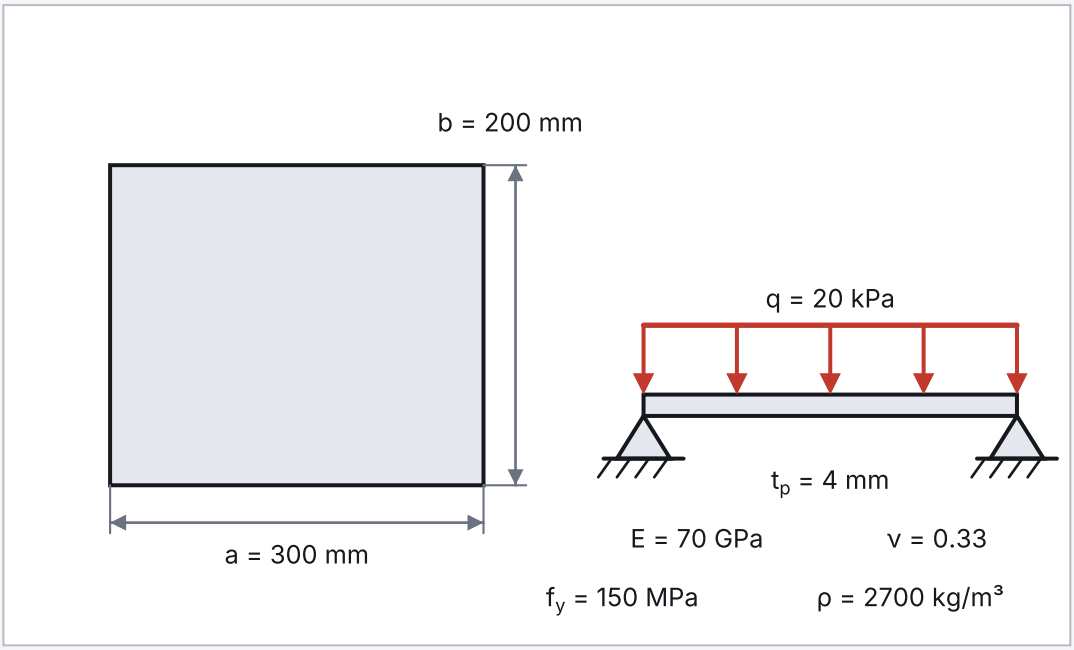


Figure 1 — A rectangular plate a by b, simply supported all round, under a uniform load given as the pressure q; in section the plate t_p on its edge supports. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Flexural rigidity	$D = D_{flex} = E{\cdot}t_p^3/(12(1-\nu^2))$; surface stress $\sigma = 6M/t_p^2$; von Mises $\sigma_{vm} = \sqrt{(\sigma_x^2 - \sigma_x{\cdot}\sigma_y + \sigma_y^2 + 3\tau_{xy}^2)}$
Circular, uniform q, clamped	$w = q{\cdot}a^4/(64D)$; $\sigma_{edge} = 3q{\cdot}a^2/(4t_p^2)$; simply supported: $w = (5+\nu){\cdot}q{\cdot}a^4/(64(1+\nu)D)$; $\sigma_{centre} = 3(3+\nu){\cdot}q{\cdot}a^2/(8t_p^2)$
Circular, central patch (radius c, pressure q) or ring (radius b, force F)	$w = A + B{\cdot}r^2$ (+ $q{\cdot}r^4/64D$) inside, $A' + B'{\cdot}r^2 + C'{\cdot}\ln(r/a) + k{\cdot}r^2{\cdot}\ln(r/a)$ outside, $k = q{\cdot}c^2/8D$ or $F/8\pi D$; constants from continuity of w, w′, M_r at the load radius and the edge conditions; $M_r = -D(w'' + \nu{\cdot}w'/r)$, $M_\theta = -D(w'/r + \nu{\cdot}w'')$
Rectangular, simply supported	$w = \Sigma\, W_{mn}{\cdot}\sin(m\pi x/a){\cdot}\sin(n\pi y/b)$ over odd m, n; $W_{mn} = q_{mn}/(\pi^4 D(m^2/a^2 + n^2/b^2)^2)$; $q_{mn} = 16q/(\pi^2 mn)$ (uniform) or $16q/(\pi^2 mn){\cdot}\sin(m\pi/2){\cdot}\sin(n\pi/2){\cdot}\sin(m\pi u/2a){\cdot}\sin(n\pi v/2b)$ (patch u × v); $M_x = D\pi^2\Sigma W_{mn}(m^2/a^2 + \nu{\cdot}n^2/b^2){\cdot}\sin{\cdot}\sin$, M_y likewise, $M_{xy} = -D(1-\nu)\pi^2\Sigma W_{mn}(mn/ab){\cdot}\cos{\cdot}\cos$; terms added until w, the centre moments and the corner M_{xy} each change by $< 10^{-6}$ over a doubling (cap 1024 × 1024; the corner sum of a uniform load, which converges like $1/N^2$, is Richardson-extrapolated before it is compared)

WHAT IT CALCULATES	RELATION
Rectangular, clamped	Ritz, $w = \Sigma\, c_{ij} \cdot X_i(x/a) \cdot Y_j(y/b)$, $X_k(\xi) = \xi^2(1-\xi)^2 \cdot P_{2k}(2\xi - 1)$; c from the minimum of the total potential energy; order K taken as 8, 12, 16, 20, 25, 30 until w and the stress change by $< 10^{-5}$ (or the change stops falling)
Fundamental frequency	simply supported rectangle $\omega_{11} = \pi^2(1/a^2 + 1/b^2) \cdot \sqrt{(D/(\rho \cdot t_p))}$, $f = \omega/2\pi$; circular $\omega = (\lambda^2/a^2) \cdot \sqrt{(D/(\rho \cdot t_p))}$, λ^2 from the Bessel characteristic equation; clamped rectangle by Rayleigh–Ritz with the same basis
Plate checks	$\sigma_{vm} \le f_y/\gamma_{M0}$; $w_{max} \le$ span/ratio (circular span = diameter, rectangle span = shorter side); WARN $w/t_p > 0.5$

Basis: Symmetrical bending of circular plates[1](Ch. 3, 2nd ed.); Simply supported rectangular plates (Navier solution)[1](Ch. 5, 2nd ed.); Rectangular plates with various edge conditions[1](Ch. 6, 2nd ed.); Natural frequencies of circular and rectangular plates[2](Ch. 2 and 4, 1st ed.); Partial factor γ_M0[3](cl. 6.1(1)); Yield criterion for the cross-section[3](cl. 6.2.1(5)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Plate shape	`rectangular`		drawing: ENC-LID-01 rev A	any
support	Edge support, all round	`simply supported`		assumption: The lid sits on a gasket land and is bolted at close pitch: treated as simply supported, which is the safe side for deflection and the central stress.	any
$load_{type}$	Load type	`uniform`		spec: Enclosure datasheet ENC-DS-07: internal pressure rise in a sealed enclosure on thermal cycling	any
$load_{basis}$	Uniform and patch loads given as	`pressure q`		assumption: Load given as a pressure.	any
a	Plate radius (circular) or length in the x direction (rectangular)	300	mm	drawing: ENC-LID-01 rev A	10 ... 10000 mm
b	Length in the y direction (rectangular plate only, not used for a circular plate)	200	mm	drawing: ENC-LID-01 rev A	10 ... 10000 mm
t_p	Plate thickness	4	mm	drawing: ENC-LID-01 rev A	0.5 ... 100 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Young's modulus	70	GPa	reference: record:5eed0001-0000-4000-8000-000000000003@1 W3 generic seed 'Aluminium alloy (generic)', E: indicative, not from a standard; replace with the material record for the alloy on the drawing	0.1 … 500 GPa
v	Poisson's ratio	0.33		reference: record:5eed0001-0000-4000-8000-000000000003@1 W3 generic seed 'Aluminium alloy (generic)', nu: indicative	0 … 0.45
ρ	Density (for the frequency)	2700	kg/m³	reference: record:5eed0001-0000-4000-8000-000000000003@1 W3 generic seed 'Aluminium alloy (generic)', rho: indicative	100 … 20000 kg/m³
f_y	Yield strength	150	MPa	assumption: Assumed 0.2 % proof strength for the sheet temper (the W3 generic seed has no strength); replace with the material record from the mill certificate.	5 … 2000 MPa
q	Pressure on the loaded area (uniform and patch loads, pressure basis)	20	kPa	spec: Enclosure datasheet ENC-DS-07: 0.2 bar differential	0.001 … 5000 kPa
F	Total force (force basis: uniform, patch and ring loads; a ring load always uses the force)	5	kN	assumption: Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.	0.001 … 10000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
r0	Patch radius (circular plate, central patch)	50	mm	assumption: Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.	1 … 5000 mm
u	Patch size in x (rectangular plate, central patch)	100	mm	assumption: Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.	1 … 10000 mm
v	Patch size in y (rectangular plate, central patch)	100	mm	assumption: Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.	1 … 10000 mm
r_{ring}	Ring radius (circular plate, ring load)	150	mm	assumption: Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.	1 … 5000 mm
γ_{M0}	Partial factor on the resistance of the cross-section	1.1		assumption: EN 1999-1-1 partial factor γ_M1 = 1.1 for the resistance of cross-sections of aluminium, entered in the γ_M0 slot; to be checked against the standard and National Annex in use	1 … 1.5
$defl_{ratio}$	Deflection limit as span/ratio (span = diameter of a circular plate, shorter side of a rectangle)	250		spec: Project specification: lid deflection limited to span/250 so the gasket stays seated	50 … 2000

3. Assumptions

A1. The plate is thin, flat, isotropic, linear elastic and of constant thickness, so Kirchhoff small-deflection theory applies. (The middle surface is unstrained, and shear deformation is neglected, so the sheet is blocked below span/t of 10.)
A2. The deflection stays small against the thickness, with w/t at most 0.5. (Above 0.5, membrane action stiffens the plate, and the small-deflection result overstates the deflection and the bending stress. The sheet warns.)
A3. The edge support is the same all round and ideal. (A simply supported edge is held at zero deflection

and is free to rotate, and a clamped edge allows no deflection or rotation. There are no elastic edges.)
A4. A simply supported rectangle has its corners held down. (The sheet reports the corner reaction $2{\cdot}M_{xy}$. If the corners can lift, the deflection and the central moments are larger than computed.)
A5. The von Mises stress is that of the plane-stress field at the plate surface, from the bending and twisting moments. (The stress is $6M/t^2$. The through-thickness shear and the local stress under a concentrated load are not included.)
A6. The stress of a clamped rectangle is scanned on a 21 × 21 grid over a quarter of the plate. (A simply supported rectangle is governed at the centre or at a corner, where the twisting moment acts.)
A7. The patches are centred and a ring load is concentric. (A load smaller than the thickness is a concentrated load, and the sheet warns (rule `patch-thin`). Patches below 1 % (circular), 2 % (simply supported rectangle) or 25 % (clamped rectangle) of the plate size are refused.)
A8. The clamped rectangle is a Ritz approximation, so its moments and stress converge more slowly than its deflection. (The sheet reports the order used and the last change, which understates the error for long plates and small patches. Compare a critical result with finite elements.)
A9. The material properties E, ν, ρ and f_y are inputs, each citing its source when taken from the material library. (The generic textbook seeds in the library are indicative.)
A10. support (Edge support, all round) is taken as simply supported. (The lid sits on a gasket land and is bolted at close pitch: treated as simply supported, which is the safe side for deflection and the central stress.)
A11. load_{basis} (Uniform and patch loads given as) is taken as pressure q. (Load given as a pressure.)
A12. f_y (Yield strength) is taken as 150 MPa. (Assumed 0.2 % proof strength for the sheet temper (the W3 generic seed has no strength); replace with the material record from the mill certificate.)
A13. F (Total force (force basis: uniform, patch and ring loads; a ring load always uses the force)) is taken as 5 kN. (Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.)
A14. r0 (Patch radius (circular plate, central patch)) is taken as 50 mm. (Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.)
A15. u (Patch size in x (rectangular plate, central patch)) is taken as 100 mm. (Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.)
A16. v (Patch size in y (rectangular plate, central patch)) is taken as 100 mm. (Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.)
A17. r_{ring} (Ring radius (circular plate, ring load)) is taken as 150 mm. (Not used: the lid carries a uniform pressure, not a force, patch or ring load; template default kept.)
A18. γ_{M0} (Partial factor on the resistance of the cross-section) is taken as 1.1. (EN 1999-1-1 partial factor γ_{M1} = 1.1 for the resistance of cross-sections of aluminium, entered in the γ_{M0} slot; to be checked against the standard and National Annex in use)

4. Working

[15] $F_{total} = q{\cdot}A_{loaded}$ [4](theory)
= 20 kPa × 60000 mm²
= 1.2 kN
Total load on the plate

[16] $D_{flex} = E{*}t_p\text{^}3/(12{*}(1 - \nu\text{^}2))$ [4](theory)
= 70 GPa × (4 mm)^3 / (12 × (1 - 0.33^2))
= 0.41896 kN·m
Flexural rigidity of the plate

[17] terms_{used} = odd terms per direction N (m, n = 1, 3, … 2N - 1) [5](5)
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 128
Terms of the Navier series used

[18] $w_{max} = 16q/(\pi^6 D){\cdot}\Sigma\ (-1)\text{^}((m+n)/2-1)/(m{\cdot}n{\cdot}(m^2/a^2 + n^2/b^2)^2)$, m, n odd [5](5)
= 128 odd terms per direction (16384 terms), last change 6.5537e-7
= 0.58996 mm

Maximum deflection (centre)

[19] $M_{x,c} = D \cdot \pi^2 \cdot \Sigma\ W_{mn}(m^2/a^2 + \nu \cdot n^2/b^2) \cdot \sin \cdot \sin$ at the centre [5](5)

= 128 odd terms per direction (16384 terms), last change 6.5537e-7

= 0.04162 kN·m/m

M_x at the centre

[20] $M_{y,c} = D \cdot \pi^2 \cdot \Sigma\ W_{mn}(n^2/b^2 + \nu \cdot m^2/a^2) \cdot \sin \cdot \sin$ at the centre [5](5)

= 128 odd terms per direction (16384 terms), last change 6.5537e-7

= 0.0656 kN·m/m

M_y at the centre

[21] $M_{xy,k} = |D(1 - \nu) \cdot \pi^2 \cdot \Sigma\ W_{mn} \cdot (mn/ab)|$ at a corner, with the $1/N^2$ tail of the sum removed by Richardson extrapolation ($s_N + (s_N - s_N/2)/3$) [5](5)

= 128 odd terms per direction (16384 terms), last change 6.5537e-7

= 0.032874 kN·m/m

Twisting moment at a corner

[22] $R_{corner} = 2 \cdot M_{xy}$ at a corner [5](5)

= 2 × 0.032874 kN·m/m

= 0.065747 kN

Corner reaction holding the corner down

[23] $M_{vm,c} = \sqrt{(M_x^2 - M_x \cdot M_y + M_y^2)}$ at the centre [6](Mises)

= M_x = 0.04162, M_y = 0.0656 kN·m/m

= 0.057492 kN·m/m

Von Mises moment at the centre

[24] $M_{vm,k} = \sqrt{3} \cdot |M_{xy}|$ at a corner [6](Mises)

= M_{xy} = 0.032874 kN·m/m

= 0.056939 kN·m/m

Von Mises moment at a corner (pure twist)

[25] $\omega_{11} = \pi^2(1/a^2 + 1/b^2) \cdot \sqrt{(D/(\rho t))}$ [5](5)

= a = 300 mm, b = 200 mm, D_{flex} = 0.41896 kN·m, ρt = 10.8 kg/m²

= 2219.8 rad/s

Angular frequency of the (1,1) mode

[26] $M_{vm} = \sqrt{(M_1^2 - M_1 \cdot M_2 + M_2^2 + 3M_{12}^2)}$ at the governing point [6](Mises)

= centre

= 0.057492 kN·m/m

Greatest von Mises moment per unit width

[27] $\sigma_{max} = 6 \cdot M_{vm}/t^2$ [4](theory)

= 6 × 0.057492 kN·m/m / (4 mm)²

= 21.56 MPa

Greatest von Mises surface stress

[28] $\sigma_{principal} = 6 \cdot M_{principal}/t^2$ [4](theory)

= 6 × 0.0656 kN·m/m / (4 mm)²

= 24.6 MPa

Greatest principal surface stress

[29] f1 = ω/2π [2]

= D_{flex} = 0.41896 kN·m, ρt = 10.8 kg/m²

= 353.29 Hz

Fundamental natural frequency (ω is the angular frequency, rad/s; f is in Hz)

[30] $w_{over,t} = w_{max}/t_p$ [4](theory)

= 0.58996 mm / 4 mm

= 0.14749

Deflection over thickness

[31] $f_{yd} = f_y/\gamma_{M0}$ [3](6.2.1(5))

= 150 MPa / 1.1

= 136.36 MPa

Design yield strength

[32] $u_{stress} = \sigma_{max} / f_{yd}$ [3](6.2.1(5))

$= 21.56\ \text{MPa} / 136.36\ \text{MPa}$

$= 0.1581$

Stress utilisation

[33] $w_{lim} = \text{span/ratio}$ [7]

$= 200\ \text{mm} / 250$

$= 0.8\ \text{mm}$

Deflection limit

[34] $u_{defl} = w_{max} / w_{lim}$ [7]

$= 0.58996\ \text{mm} / 0.8\ \text{mm}$

$= 0.73745$

Deflection utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
w_{max}	0.5900 mm	Maximum deflection (governing)
σ_{max}	21.56 MPa	Greatest von Mises stress
f1	353.29 Hz	Fundamental frequency
D_{flex}	0.4190 kN·m	Flexural rigidity
$\sigma_{principal}$	24.60 MPa	Greatest principal stress
f_{yd}	136.4 MPa	Design yield strength
$w_{over,t}$	0.14749	Deflection over thickness
$terms_{used}$	128	Series terms used
$M_{x,c}$	0.04162 kN·m/m	M_x at the centre
$M_{y,c}$	0.06560 kN·m/m	M_y at the centre
$M_{xy,k}$	0.03287 kN·m/m	Corner twisting moment
R_{corner}	0.06575 kN	Corner reaction
location	"centre"	Governing stress point
u_{stress}	0.16	Stress utilisation
u_{defl}	0.74	Deflection utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Deflection against the deflection limit	0.5900 mm	0.8000 mm	0.74	[7]
PASS	Von Mises stress against the design yield strength	21.56 MPa	136.4 MPa	0.16	[8](6.2.1(5))

7. Applicability

WARNING records: tabulated values not taken from a reference record are f_y (Material properties); γ_{M0} (EN 1993 National Annex factors); $defl_{ratio}$ (Deflection limit). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template struct.plate-bending@1.4.0 Content hash 73b2e646dac2f675 Engine calc-engine/1.0.0
Known-answer cases passed 11/11 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Timoshenko & Woinowsky-Krieger, Theory of Plates and Shells.
2. Leissa, Vibration of Plates (NASA SP-160).
3. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
4. Kirchhoff.
5. T&WK Ch.
6. von.
7. Serviceability.
8. cl.